Curriculum Vitae/Resume

Liuyang Cheng
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Research Profile

Ph.D. candidate in Mechanical Science and Engineering, University of Illinois Urbana-Champaign (expected Dec. 2026). Five peer-reviewed journal articles, including a first-author paper in PNAS, spanning theory, manufacturing, and characterization of soft actuators.

Research Interests

Soft robotics; Bioinspired mechanism; Material actuator; Nonlinear mechanics of J-curve materials; Scalable actuator manufacturing and characterization.

Education

Ph.D. of Mechanical Science and Engineering

University of Illinois Urbana-Champaign | Illinois, USA *Aug. 2021 - Dec. 2026 (expected)*

- Thesis: Bioinspired Functional Materials and Versatile Soft Robotic Arm; Advisor: Sameh H. Tawfick
- GPA (Overall): 3.8/4.0

B.S. in Mechanical Science and Engineering

University of Illinois Urbana-Champaign | Illinois, USA *Aug. 2017 - June 2021*

- GPA (Overall): 3.72/4.0
- Dual-degree program with Zhejiang University

B.Eng. in Mechanical Engineering

Zhejiang University | Zhejiang, China *Sep. 2017 - June 2021*

- GPA (Overall): 3.75/4.0

Publications

Peer-Reviewed Journal Articles

- **Cheng, L.**, Eom, W., Wang, Q., Kim, H., Pineda Guzman, R., Kim, J., Solis, M., Malladi, S., Tsai, S., Kersh, M. E., & Tawfick, S. (2026). Principles of use of tensile J-curve materials in antagonistic arrangements. *Proceedings of the National Academy of Sciences*, 123(46), e2604227123. https://doi.org/10.1073/pnas.2604227123
- Wang, Q., **Cheng, L.**, Kim, J., Tsai, S., Roach, D., & Tawfick, S. (2026). Characterization and selection of contractile actuators for soft robotics operating in the quasistatic regime. *Advanced Intelligent Systems*, Article e70490. https://doi.org/10.1002/aisy.70490
- Tsai, S., **Cheng, L.**, Albazroun, A., Wang, Q., Kim, J., Tekinalp, A., ... & Tawfick, S. (2026). Hierarchical Artificial Muscle with Nonlinear Elasticity for Antagonistic and Cyclic Robotics. *Advanced Science*, 13(30), e21604.
- Wang, Q., Ghrayeb, A., Kim, S., **Cheng, L.**, & Tawfick, S. (2024). The mechanics and physics of twisted and coiled polymer actuators. *International Journal of Mechanical Sciences*, 280, 109440.

- Wang, Y., Wang, Q., Liu, M., Qin, Y., **Cheng, L.**, Bolmin, O., ... & Tawfick, S. (2023). Insect-scale jumping robots enabled by a dynamic buckling cascade. *Proceedings of the National Academy of Sciences*, 120(5), e2210651120.

In Preparation

- Kim, J., Wang, Q., **Cheng, L.**, Tsai, S., Kim, S.H., & Tawfick, S., "A Predictive Design Framework for a Soft Robotic Ventricle using Contractile Actuators.", [Preprint at arXiv:2609.09706].

Conference Presentations

- **Liuyang Cheng** (presenter). "Soft Spiral Octopus Tentacles Driven by Concentric Coiled Artificial Muscles." 2026 *Society of Engineering Science* Annual Technical Meeting (SES 2026), Purdue University, West Lafayette, Indiana, USA, Oct. 12, 2026. (Oral)
- **Cheng, L.**, Kim, J., & Tawfick, S., "Soft Robotic Octopus Tentacles Actuated by Scalable Coiled Artificial Muscles." *Smart Materials, Adaptive Structures, and Intelligent Systems*, St. Louis, Missouri, Sept. 2025. (Oral)
- **Cheng, L.**, Eom, W., Wang, Q., Kersh, M. E., & Tawfick, S. "Hierarchical twisted and coiled fibers mimicking the stretchability and energy absorption of biological ligaments." *Society of Engineering Science*, Minneapolis, MN, Oct. 2023. (Oral)

Awards & Honors

Undergraduate Technology Innovation Award

Government of Zhejiang Province, China *Aug. 2020*

Leader of the Project Multi-functional Portable Egg Boiler & Peeler

- Top 7% student independent research projects of all universities in Zhejiang Province

Academic Service

Manuscript Reviewer, *Advanced Robotics Research* *2026*

Research Projects

Scalable Supercoiled Artificial Muscle Manufacturing

University of Illinois Urbana-Champaign *June 2023 - On-going*

- Originated a continuous fabrication method and built an Arduino-based machine for scalable production
- Developed theory linking stepper-motor speeds to twist density, geometry, and actuation performance

Octopus-Inspired Soft Robotic Arm

Mini-MURI Project, UIUC (ONR-funded) *Jan. 2023 - On-going*

Advisors: Sameh H. Tawfick, Mattia Gazzola, James Pikul

- Designed the tapered spiral geometry and built 2D prototypes with longitudinal and oblique muscles, demonstrating curling and twisting
- Established theory relating arm geometry to the required muscle stroke
- Added a mesh structure for greater flexibility; building a 3D spiral version

Actuator Characterization and Selection for Soft Robotic Design

University of Illinois Urbana-Champaign *July 2024 - Oct. 2025*

- Developed a two-tensile-test method for actuator selection, validated on McKibben actuators

Smart Granular Robot with Vibration Sensing

University of Illinois Urbana-Champaign *June 2023 – Mar. 2025*

Advisors: Sameh H. Tawfick, Ahmed Elbanna

- Developed a granular robot with vibration sensing, an on-board micro-motor, and inter-robot communication
- Built a Python interface for group control; evaluated environmental interaction as a smart soil

Principles of Use of Tensile J-Curve Materials in Antagonistic Arrangements

University of Illinois Urbana-Champaign *May 2022 - April 2025*

- Led the design of artificial ligaments reproducing the J-shaped tensile response of biological ligaments; identified the mechanisms of their nonlinearity and hysteresis
- Originated principles of use of J-curve materials in linear and rotary antagonistic mechanisms (first-author paper published in PNAS, 2026)
- Developed self-sensing ligaments using CNT surface coatings

Supercoiled & Hypercoiled Artificial Muscles

University of Illinois Urbana-Champaign *Feb. 2022 - May 2024*

- Led the design of supercoiled and hypercoiled artificial muscles with embedded nichrome wires for uniform Joule heating; built an Arduino-based fabrication machine
- Helped standardize isobaric and isometric testing methods

Insect-Scale Jumping Robot with Coiled Artificial Muscle

University of Illinois Urbana-Champaign *June 2020 - May 2022*

- Fabricated coiled muscles and tested the insect-scale jumping robot (PNAS, 2023)

High Power Density, Large-Stroke Coiled Artificial Muscle System

Undergraduate Research Assistant, UIUC *Jan. 2020 - Aug. 2020*

- Optimized twisting and mandrel-coiling procedures across five approaches; quantified effects of material, diameter, and pre-stretch
- Standardized fabrication and produced custom muscles for a collaborating group at Harvard

Teaching Experience

TAM 210/211: Engineering Mechanics – Statics
University of Illinois Urbana-Champaign *Aug. 2025 - Dec. 2025*
Teaching Assistant

ME 170: Computer-Aided Design
University of Illinois Urbana-Champaign *Jan. 2025 - June 2025*
Teaching Assistant

ME 370: Mechanical Design I
University of Illinois Urbana-Champaign *Jan. 2024 - June 2024*
Teaching Assistant

TAM 211: Engineering Mechanics – Statics
Zhejiang University *Sep. 2019 - Jan. 2020*
Teaching Assistant

Technical Skills

- **Fabrication:** twisted and coiled polymer actuators (mono-, super- and hyper-coiled), McKibben actuators, embedded nichrome Joule-heating, SIS fiber spinning, mandrel coiling, scalable muscle manufacturing
- **Characterization:** isobaric and isometric actuator testing, tensile and cyclic testing, two-tensile-test actuator selection, J-curve nonlinearity and hysteresis analysis
- **Design and computation:** CAD modeling, Arduino-based machine design and stepper-motor control, soft arm and mesh assembly, MATLAB, Python, machine learning